

```
In [290]: #Importing the required libraries  
import pandas as pd  
import numpy as np  
%matplotlib inline  
import matplotlib as mpl  
mpl.style.use("ggplot")  
import matplotlib.pyplot as plt  
import scipy as sp  
import sklearn
```

Here we use the flights data and the planes data from R under the library "nycflights13" to perform our exploratory analysis

```
In [291]: #Read the csv file into a local variable to create a data frame  
flights_df= pd.read_csv('flights.csv')
```

In [292]: *#Use the head function to see the first 10 rows of the data frame*
 flights_df.head(10)

Out[292]:

	Unnamed: 0	year	month	day	dep_time	dep_delay	arr_time	arr_delay	carrier	tailnum	flight	origin	dest	air_time	dist
0	1	2013	1	1	517	2	830	11	UA	N14228	1545	EWR	IAH	227	1400
1	2	2013	1	1	533	4	850	20	UA	N24211	1714	LGA	IAH	227	1416
2	3	2013	1	1	542	2	923	33	AA	N619AA	1141	JFK	MIA	160	1085
3	4	2013	1	1	544	-1	1004	-18	B6	N804JB	725	JFK	BQN	183	1576
4	5	2013	1	1	554	-6	812	-25	DL	N668DN	461	LGA	ATL	116	762
5	6	2013	1	1	554	-4	740	12	UA	N39463	1696	EWR	ORD	150	719
6	7	2013	1	1	555	-5	913	19	B6	N516JB	507	EWR	FLL	158	1065
7	8	2013	1	1	557	-3	709	-14	EV	N829AS	5708	LGA	IAD	53	229
8	9	2013	1	1	557	-3	838	-8	B6	N593JB	79	JFK	MCO	140	944
9	10	2013	1	1	558	-2	753	8	AA	N3ALAA	301	LGA	ORD	138	733

In [293]: *#Check to see all the destination airports*
 flights_df["dest"].unique()

Out[293]: array(['IAH', 'MIA', 'BQN', 'ATL', 'ORD', 'FLL', 'IAD', 'MCO', 'PBI',
 'TPA', 'LAX', 'SFO', 'DFW', 'BOS', 'LAS', 'MSP', 'DTW', 'RSW',
 'SJU', 'PHX', 'BWI', 'CLT', 'BUF', 'DEN', 'SNA', 'MSY', 'SLC',
 'XNA', 'MKE', 'SEA', 'ROC', 'SYR', 'SRQ', 'RDU', 'CMH', 'JAX',
 'CHS', 'MEM', 'PIT', 'SAN', 'DCA', 'CLE', 'STL', 'MYR', 'JAC',
 'MDW', 'HNL', 'BNA', 'AUS', 'BTV', 'PHL', 'STT', 'EGE', 'AVL',
 'PWM', 'IND', 'SAV', 'CAK', 'HOU', 'LGB', 'DAY', 'ALB', 'BDL',
 'MHT', 'MSN', 'GSO', 'CVG', 'BUR', 'RIC', 'GSP', 'GRR', 'MCI',
 'ORF', 'SAT', 'SDF', 'PDX', 'SJC', 'OMA', 'CRW', 'OAK', 'SMF',
 'TUL', 'TYS', 'OKC', 'PVD', 'DSM', 'PSE', 'BHM', 'CAE', 'HDN',
 'BZN', 'MTJ', 'EYW', 'PSP', 'ACK', 'BGR', 'ABQ', 'ILM', 'MVY',
 'SBN', 'LEX', 'CHO', 'TVC', 'ANC', 'LGA'], dtype=object)

```

In [312]: flights_df.head()
#Store the data frame in a new variable
flights = flights_df

#Create a new column in the data frame called dates
flights["date"] = pd.to_datetime(flights['year']*10000 + flights['month']*100 + flights['day'], format="%Y%m%d")
#Group all the rows by dates and find mean of the other column values
flights_new = flights.reset_index().groupby(['date']).mean()

```

```

In [313]: flights_new.head()

```

```

Out[313]:

```

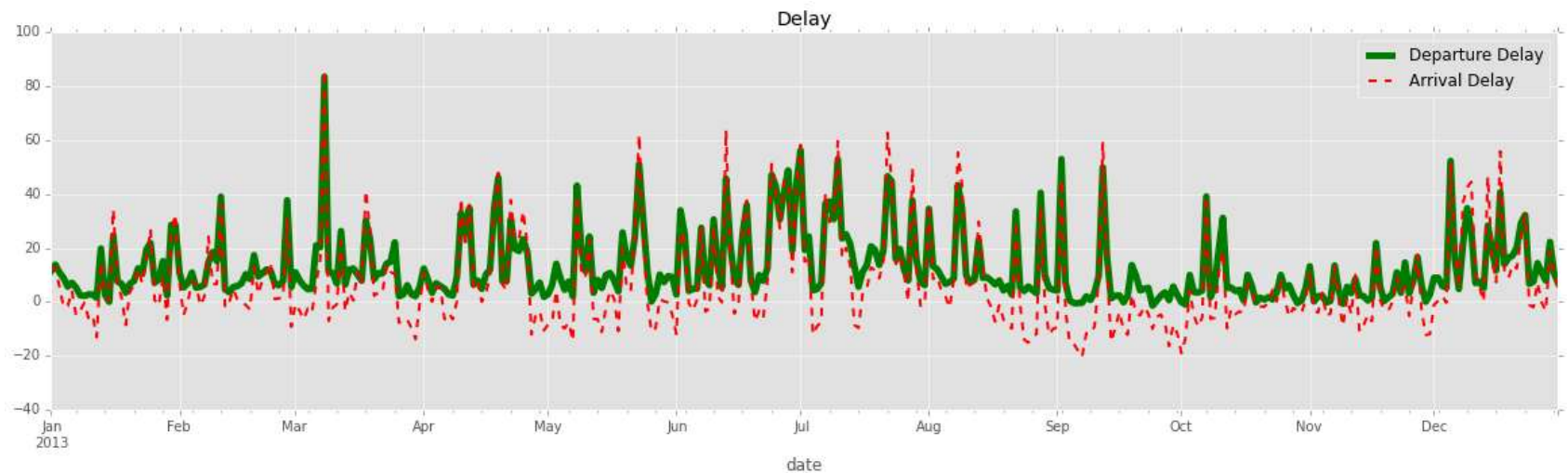
	index	Unnamed: 0	year	month	day	dep_time	dep_delay	arr_time	arr_delay	flight	air_time	dista
date												
2013-01-01	420.5	421.5	2013	1	1	1384.991647	11.548926	1562.344086	12.651023	1821.496437	169.652226	1077
2013-01-02	1313.0	1314.0	2013	1	2	1353.960428	13.858824	1532.729904	12.692888	1917.732768	162.198276	1053
2013-01-03	2241.5	2242.5	2013	1	3	1356.665929	10.987832	1536.290929	5.733333	1913.176149	156.593333	1037
2013-01-04	3156.0	3157.0	2013	1	4	1347.856986	8.951595	1519.116612	-1.932819	1930.147541	150.997797	1032
2013-01-05	3973.5	3974.5	2013	1	5	1326.089261	5.732218	1509.136681	-1.525802	1740.019444	160.792190	1067

```
In [314]: #Plot showcasing the departure delays for the year 2013
flights_new["dep_delay"].plot(figsize=(20, 5),color='green', title='Delay', linewidth = 5, label = "Departure Delay")

flights_new["arr_delay"].plot(figsize=(20, 5),color='red', title='Delay',\
                               linestyle = "dashed", linewidth = 2, label = "Arrival Delay")

import pylab
pylab.legend(loc = "upper right")
#plt.plot(flights["dep_delay"],color="red",linewidth = 1.5)
#plt.text(70, 80, r'8th March')
#plt.title("Worst day to fly out of NYC")

plt.show()
```



We can see from the above plot that the highest departure and arrival delay was some time in the start of March which we calculated and got as 8th March. This date was definitely the worst date to fly on for 2013.

```
In [299]: #Create a new data frame which are grouped by destination and we take the mean of the other columns
          flights_dest = flights_df.groupby('dest').mean()
          #lights_explore.head(20)

          #Create a new column called dest and use the index to populate the values
          flights_dest["dest"] = flights_dest.index
          flights_dest.head(10)
```

Out[299]:

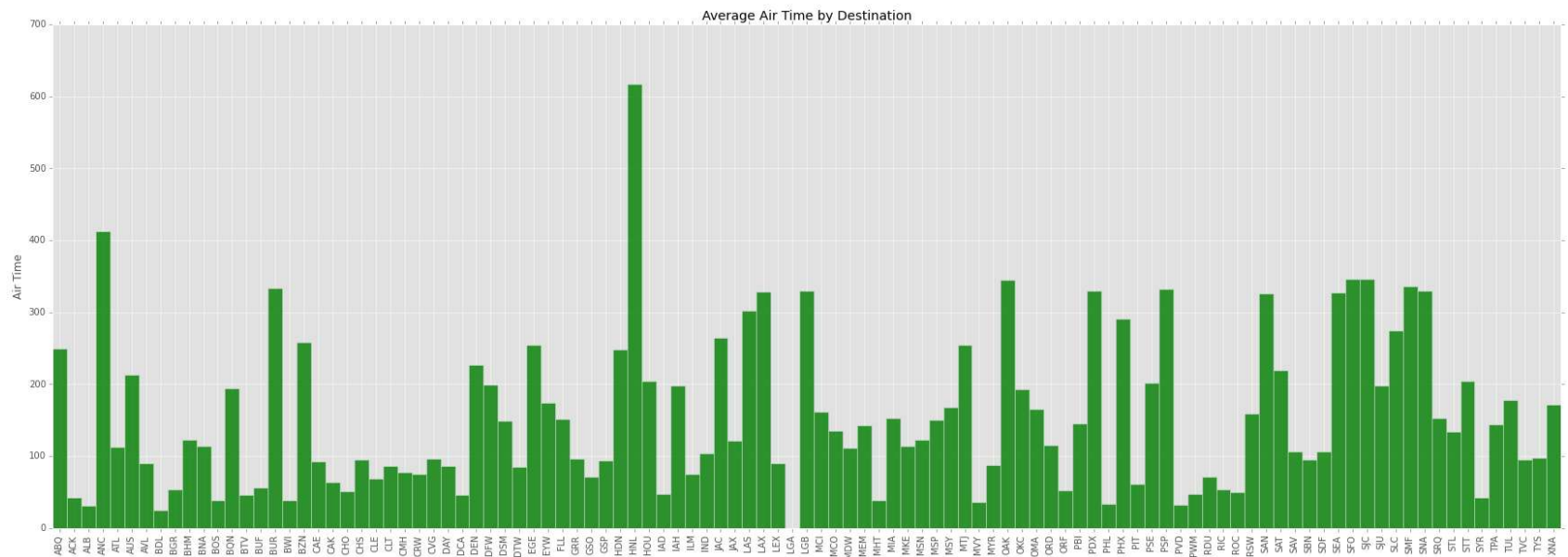
	Unnamed: 0	year	month	day	dep_time	dep_delay	arr_time	arr_delay	flight	air_time
dest										
ABQ	191964.271654	2013	8.338583	16.177165	2005.732283	13.740157	2049.397638	4.381890	830.354331	249.169291
ACK	254071.362264	2013	7.532075	16.381132	1032.664151	6.456604	1145.079245	4.852273	1301.641509	42.068182
ALB	145174.808656	2013	5.665148	15.136674	1627.188544	23.620525	1701.593301	14.397129	4561.744875	31.787081
ANC	279033.000000	2013	7.500000	15.000000	1635.375000	12.875000	1968.000000	-2.500000	887.000000	413.125000
ATL	168184.380076	2013	6.536509	15.712286	1293.291395	12.509824	1513.462336	11.300113	1881.501481	112.930451
AUS	170830.947109	2013	6.488725	15.679787	1521.476427	13.025641	1614.333058	6.019909	1024.496515	212.727914
AVL	196533.741818	2013	8.170909	15.716364	1174.764259	8.190114	1372.638783	8.003831	4278.534545	89.888889
BDL	182838.677201	2013	5.345372	15.758465	1490.310680	17.720874	1548.604369	7.048544	4353.214447	25.466019
BGR	173297.730667	2013	8.709333	16.112000	1689.716667	19.475000	1715.203911	8.027933	5299.541333	54.117318
BHM	175548.336700	2013	6.228956	15.949495	1943.639706	29.694853	2028.096654	16.877323	5033.511785	122.776952

```
In [300]: #Create a plot that shows the average air time based on destination
y_pos = np.arange(len(flights_dest["dest"]))
performance = flights_dest["air_time"]

plt.figure(figsize=(30,10))

plt.bar(y_pos, performance, alpha=0.8, color = "green",width=1, align = "center")
plt.xticks(y_pos, flights_dest["dest"],rotation=90)
plt.ylabel('Air Time')
plt.title('Average Air Time by Destination')

plt.autoscale()
plt.show()
```



From this plot we can see that the flights to Honolulu were on average the longest "in air" flights from NYC

```
In [301]: #Create a new data frame which are grouped by destination and we take the mean of the other columns
flights_dest = flights_df.groupby('dest').sum()
#lights_explore.head(20)

#Create a new column called dest and use the index to populate the values
flights_dest["dest"] = flights_dest.index
flights_dest.head(10)
```

Out[301]:

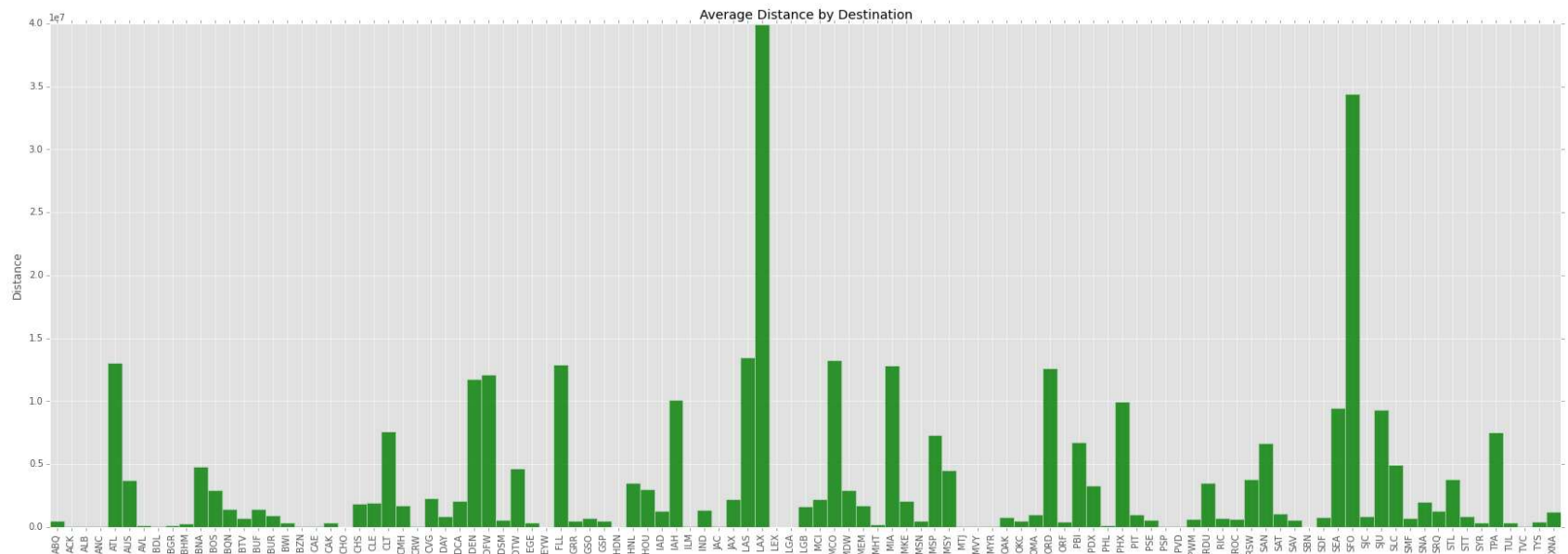
	Unnamed: 0	year	month	day	dep_time	dep_delay	arr_time	arr_delay	flight	air_time	distance	hour
dest												
ABQ	48758925	511302	2118	4109	509456	3490	520547	1113	210910	63289	463804	5013
ACK	67328911	533445	1996	4341	273656	1711	303446	1281	344935	11106	52735	2646
ALB	63731741	883707	2487	6645	681792	9897	711266	6018	2002606	13287	62777	6696
ANC	2232264	16104	60	120	13083	103	15744	-20	7096	3305	26960	129
ATL	2895294103	34653795	112526	270487	21854038	211391	25536650	190260	32390048	1901410	13033618	2126
AUS	416656680	4909707	15826	38243	3678930	31496	3901843	14514	2498747	512887	3693263	3606
AVL	54046779	553575	2247	4322	308963	2154	361004	2089	1176597	23461	160485	3002
BDL	80997534	891759	2368	6981	614008	7301	638025	2904	1928474	10492	51388	6019
BGR	64986649	754875	3266	6042	608298	7011	614043	2874	1987328	19374	141750	5966
BHM	52137856	597861	1850	4737	528670	8077	545558	4540	1494953	33027	257201	5194

```
In [333]: #Create a plot that shows the average air time based on destination
y_pos = np.arange(len(flights_dest["dest"]))
performance = flights_dest["distance"]

plt.figure(figsize=(30,10))

plt.bar(y_pos, performance, alpha=0.8, color = "green",width=1, align = "center")
plt.xticks(y_pos, flights_dest["dest"],rotation=90)
plt.ylabel('Distance')
plt.title('Average Distance by Destination')

plt.autoscale()
plt.show()
```



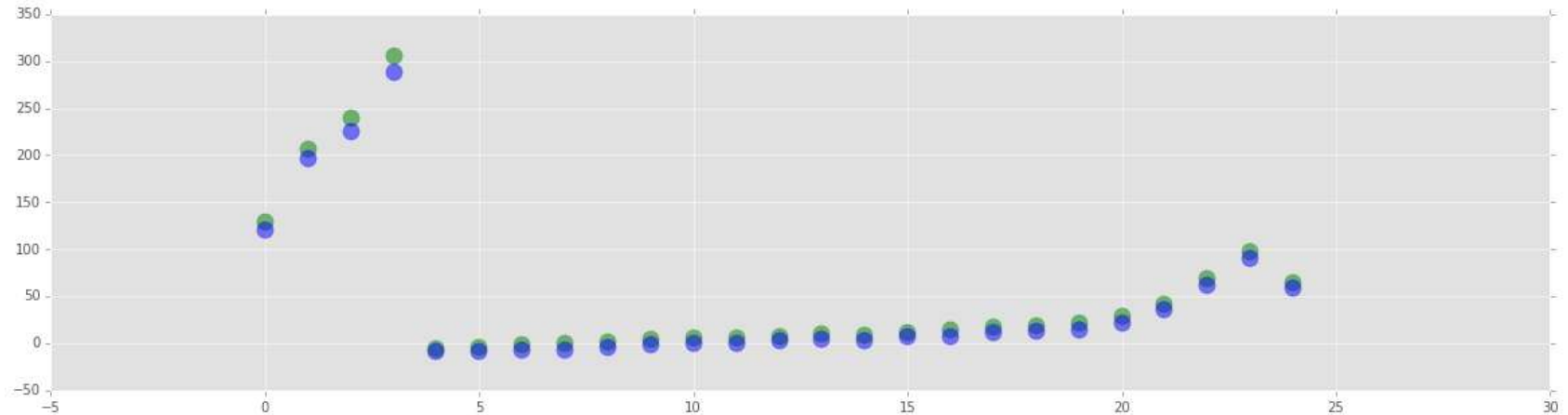
This plot shows that on average the distance covered by flights from NYC to Los Angeles were the longest in terms of distance which is understandable since the two locations are on the opposite ends of the country.


```
In [331]: flights_new_hour["hour"] = flights_new_hour.index
flights_new_hour.head(4)

plt.figure(figsize=(20,5))

plt.scatter(flights_new_hour["hour"], flights_new_hour["dep_delay"], s=150, alpha=0.5, color = "green")
plt.scatter(flights_new_hour["hour"], flights_new_hour["arr_delay"], s=150, alpha=0.5, color = "blue")

Out[331]: <matplotlib.collections.PathCollection at 0x12d2646d0>
```

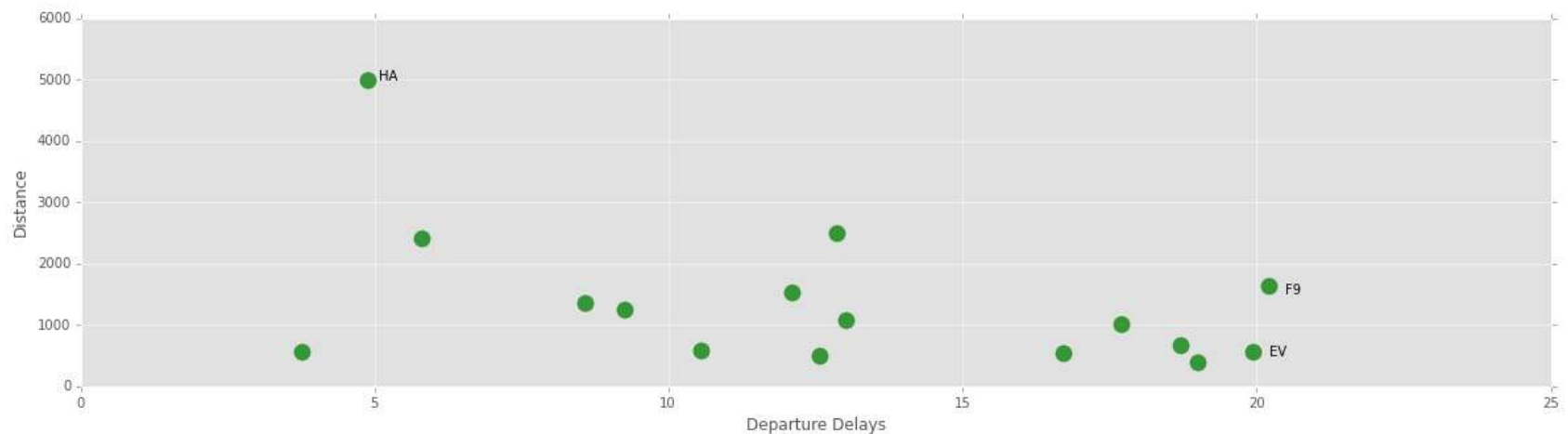


As we can clearly see that the departure delay is at its highest during the early morning hours from midnight to 4AM. Through the rest of the day although there is some departure delay, it is not as significant as the early morning delays. Also, we can see that at around 4-5AM the delay is minimum and tends to increase over the course of the day with the highest peak at around 3AM. From about 9PM the delays tend to start getting higher. The arrival delay follows a similar pattern as the departure delay as can be seen which might lead us to believe that the flights that arrived late were probably the ones that departed late as well. To confirm this we would have to perform further analysis.

```
In [304]: #Create a scatter plot to further showcase the relation between the Average Departure Delay and
#Average Distance travelled by the flights
plt.figure(figsize=(20,5))

colors = np.arange(len(flights_explore["carrier"]))
plt.scatter(flights_explore["dep_delay"], flights_explore["distance"], s=150, alpha=0.75, color = "green")
plt.xlabel('Departure Delays')
plt.ylabel('Distance')
plt.text(5.1, 5000, r'HA')
plt.text(20.5, 1500, r'F9')
plt.text(20.25, 500, r'EV')
#plt.legend()

plt.show()
```



We can see from the above scatter plot which is also supported by the bar plots that the flights that travel large distances tend to have lower departure delay as can be seen by the carrier "HA". We also notice a small cluster of carriers that have a higher departure delay on average and the interesting thing to note here is that they all travelled smaller distances on average, in fact, they all travelled less than 2000 miles. There are carriers where even for smaller distances there is a small delay but for the most of the carriers we can see that the average departure delay is lower if the average distance is higher and for a higher departure delay we have carriers travelling a smaller distance on average.

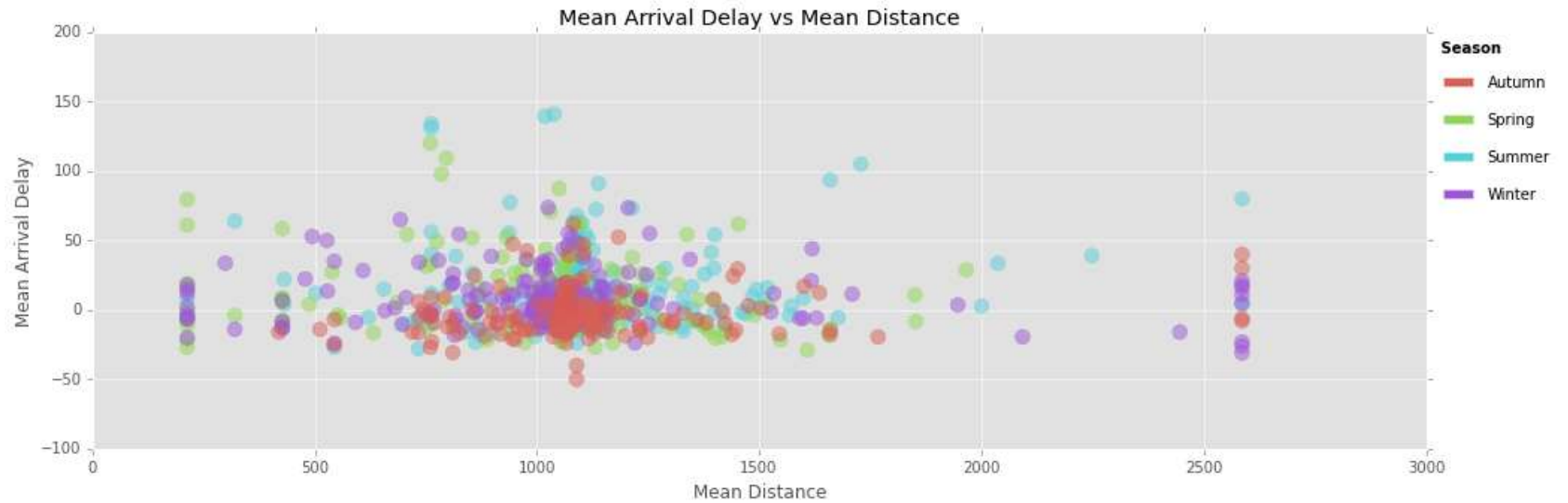
```
In [305]: #Store the planes data set in a new data frame
planes_df= pd.read_csv('planes.csv')

#Merge the two data frames flights and planes together by using left join based on the tailnum column
flights_planes = pd.merge(flights, planes_df, how='left', on=['tailnum'])
#planes_df.head(10)

#Group the data in the new merged data frame based on date and engines and then take the mean of the other columns
flights_planes_grouped = flights_planes.groupby(['date','engines']).mean()

#Create a new column called season based on the below conditions
flights_planes_grouped.loc[(flights_planes_grouped["month"] == 1) | \
                           (flights_planes_grouped["month"] == 2) | \
                           (flights_planes_grouped["month"] == 12),"season"]="Winter"
flights_planes_grouped.loc[(flights_planes_grouped['month'] == 3) | \
                           (flights_planes_grouped['month'] == 4) | \
                           (flights_planes_grouped['month'] == 5), 'season'] = 'Spring'
flights_planes_grouped.loc[(flights_planes_grouped['month'] == 6) | \
                           (flights_planes_grouped['month'] == 7) | \
                           (flights_planes_grouped['month'] == 8), 'season'] = 'Summer'
flights_planes_grouped.loc[(flights_planes_grouped['month'] == 9) | \
                           (flights_planes_grouped['month'] == 10) | \
                           (flights_planes_grouped['month'] == 11), 'season'] = 'Autumn'

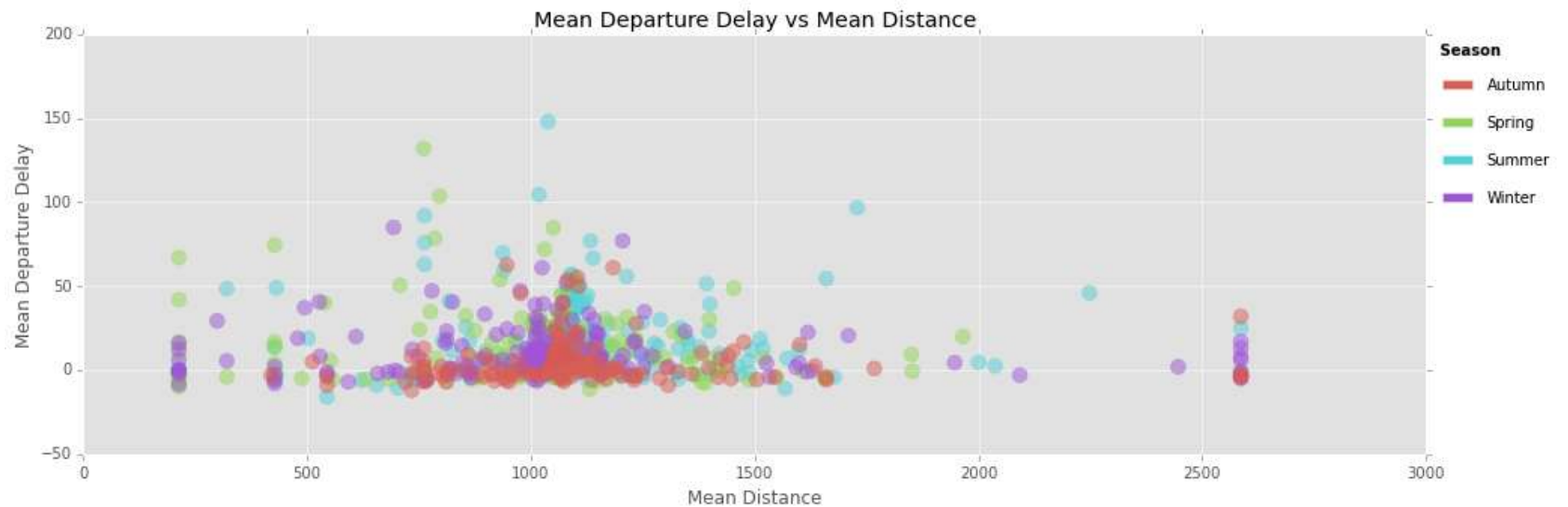
flights_planes_grouped.reset_index(inplace=True)
flights_planes_grouped.head(20)
#flights.head()
#Plot a relation between arrival delay and distance
gsize = theme_matplotlib(rc={"figure.figsize": "20, 5"}, matplotlib_defaults=False)
ggplot(aes(x="distance",y="arr_delay",color="season"),data=flights_planes_grouped)+\
geom_point(size = 100, alpha=0.5)+\
xlab("Mean Distance")+ylab("Mean Arrival Delay")+ggtitle("Mean Arrival Delay vs Mean Distance")+gsize
```



```
Out[305]: <ggplot: (331524537)>
```

From the above plot we can see that most of the flights have a mean distance ranging from 750 to 1500 miles. There are a few outliers as well that are beyond 2500 miles but what is the most noticeable in the plot is the cluster of distances during Autumn. Also, most of the arrival delays seem to happen during summer.

```
In [306]: flights_planes_grouped.reset_index(inplace=True)
flights_planes_grouped.head(20)
#flights.head()
#Plot a relation between departure delay and distance
gsize = theme_matplotlib(rc={"figure.figsize": "20, 5"}, matplotlib_defaults=False)
ggplot(aes(x="distance",y="dep_delay",color="season"),data=flights_planes_grouped)+\
geom_point(size = 100, alpha = 0.5)+xlab("Mean Distance")+ylab("Mean Departure Delay")+\\
ggtitle("Mean Departure Delay vs Mean Distance")+gsize
```



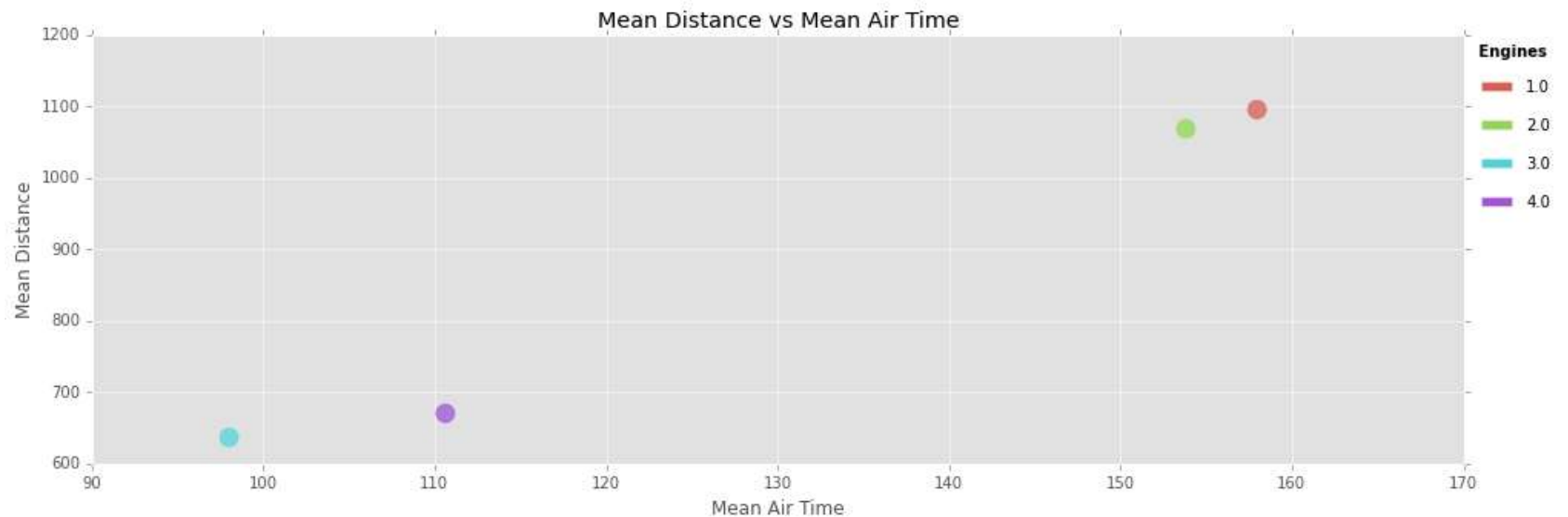
```
Out[306]: <ggplot: (331626329)>
```

From the above plot we can see a most of the points are clustered between 700 to 1500 miles and similar to the plot comprising of the arrival delay we can see some outliers beyond the 2500 miles mark. There are 3-4 cases of where the departure delay is high during the summer and 1 such case during spring.

```
In [307]: #Group the data in the new merged data frame based on engines and take the mean of the other columns
flights_planes_grouped = flights_planes.groupby('engines').mean()

flights_planes_grouped["engines"] = flights_planes_grouped.index
flights_planes_grouped.head(10)

gsize = theme_matplotlib(rc={"figure.figsize": "20, 5"}, matplotlib_defaults=False)
ggplot(aes(x="air_time",y="distance",color="engines"),data=flights_planes_grouped)+\
geom_point(size = 150, alpha = 0.75)+xlab("Mean Air Time")+ylab("Mean Distance")+\\
ggtitle("Mean Distance vs Mean Air Time")+gsize
```



```
Out[307]: <ggplot: (316538169)>
```

Here we try to look at the mean distance versus the mean air time. It is obvious that with increase in distance the air time is increasing. Something to note here is that air time for a 1 engine plane is highest on average. This could be because there are a lot of flights that are travelling using a single engine as compared to a flight with more number of engines.

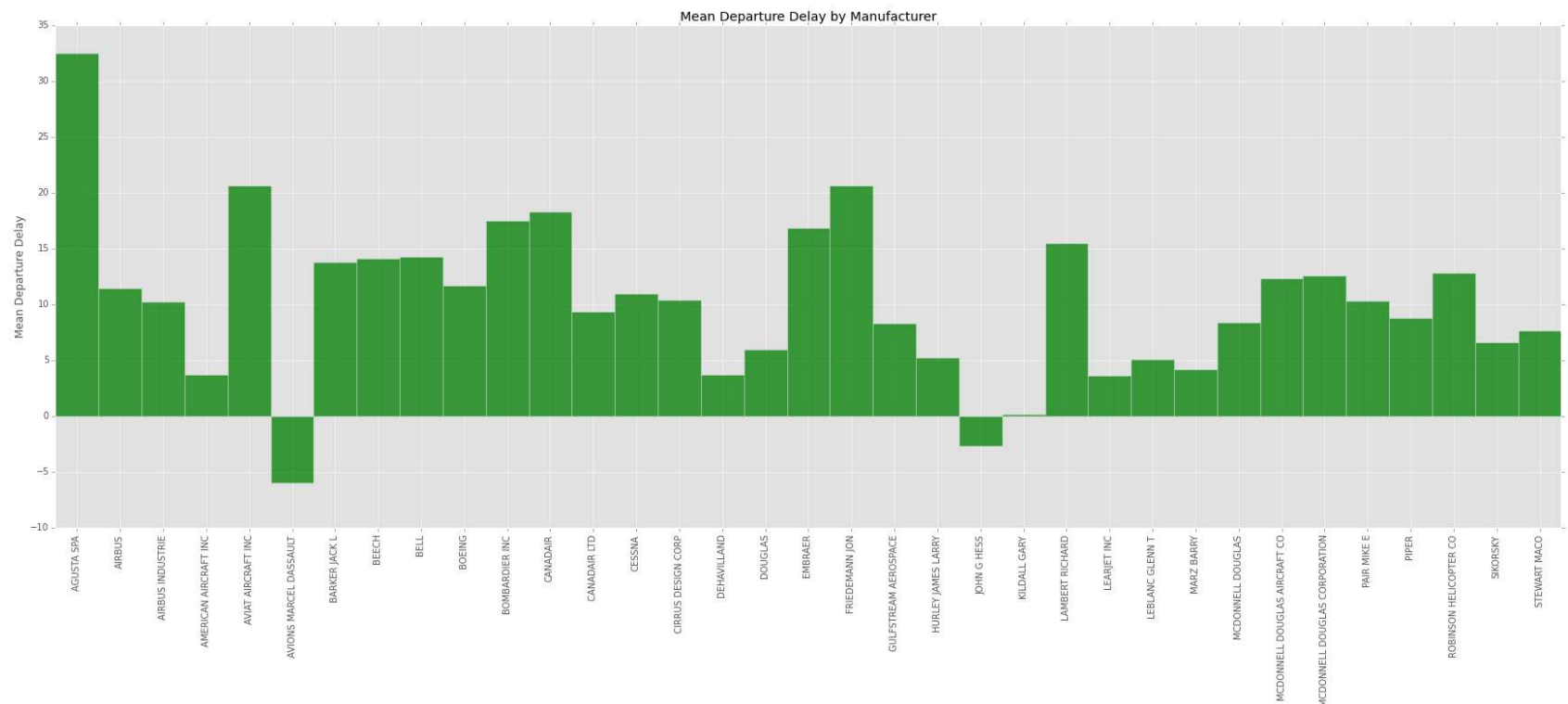
```
In [308]: #Group the data in the new merged data frame based on manufacturer and take the mean of the other columns
flights_planes_grouped = flights_planes.groupby('manufacturer').mean()
flights_planes_grouped["manufacturer"] = flights_planes_grouped.index

#Create a plot that shows the average departure delay based on manufacturer
y_pos = np.arange(len(flights_planes_grouped["manufacturer"]))
performance = flights_planes_grouped["dep_delay"]

plt.figure(figsize=(30,10))

plt.bar(y_pos, performance, alpha=0.75, color = "green",width=1, align = "center")
plt.xticks(y_pos, flights_planes_grouped["manufacturer"],rotation=90)
plt.ylabel('Mean Departure Delay')
plt.title('Mean Departure Delay by Manufacturer')

plt.autoscale()
plt.show()
```



The above plot shows that on average the departure delay is high for most of the manufacturers except 3. However, this could be owing to the fact that the flights that are manufactured by them are far greater in number. The highest departure delay is for the manufacturer "AUGUSTA SPA" which is significantly higher than the rest of the cases of departure delay.

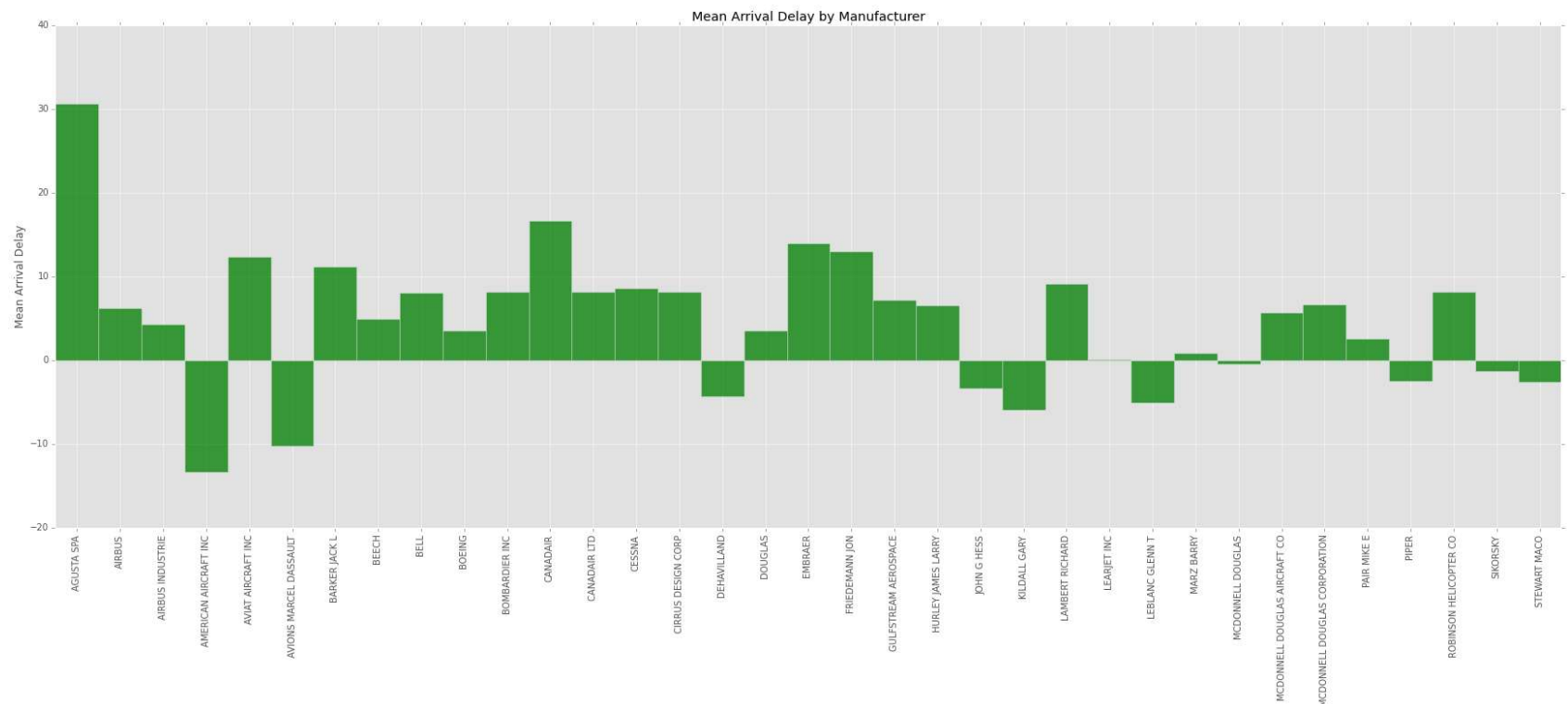

```
In [309]: #Group the data in the new merged data frame based on date and take the mean of the other columns
flights_planes_grouped = flights_planes.groupby('manufacturer').mean()
flights_planes_grouped["manufacturer"] = flights_planes_grouped.index

#Create a plot that shows the average air time based on destination
y_pos = np.arange(len(flights_planes_grouped["manufacturer"]))
performance = flights_planes_grouped["arr_delay"]

plt.figure(figsize=(30,10))

plt.bar(y_pos, performance, alpha=0.75, color = "green",width=1, align = "center")
plt.xticks(y_pos, flights_planes_grouped["manufacturer"],rotation=90)
plt.ylabel('Mean Arrival Delay')
plt.title('Mean Arrival Delay by Manufacturer')

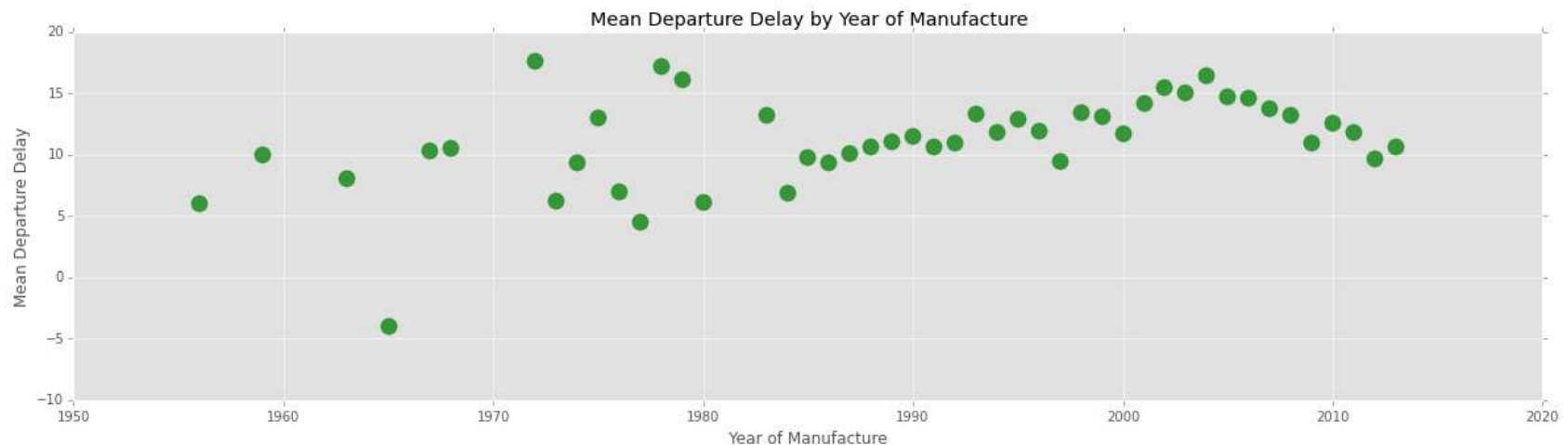
plt.autoscale()
plt.show()
```



The above plot shows that on average the arrival delay is lesser as compared to the average departure delay that we discussed about in the above plot. The arrival delay is considerably lesser for most of the manufacturers except for "AUGUSTA SPA" which has a relatively higher delay as compared to the rest of the manufacturers.

```
In [310]: #Group the data in the new merged data frame based on manufactured year and take the mean of the other columns
flights_planes_grouped = flights_planes.groupby('year_y').mean()
flights_planes_grouped["year_y"] = flights_planes_grouped.index
flights_planes_grouped.head(5)

#Create a plot that shows mean departure delay by year of manufacture
gsize = theme_matplotlib(rc={"figure.figsize": "20, 5"}, matplotlib_defaults=False)
ggplot(aes(x="year_y",y="dep_delay"),data=flights_planes_grouped)+\
geom_point(color="green", size = 150, alpha = 0.75)+xlab("Year of Manufacture")+\\
ylab("Mean Departure Delay")+ggtitle("Mean Departure Delay by Year of Manufacture")+gsize
```

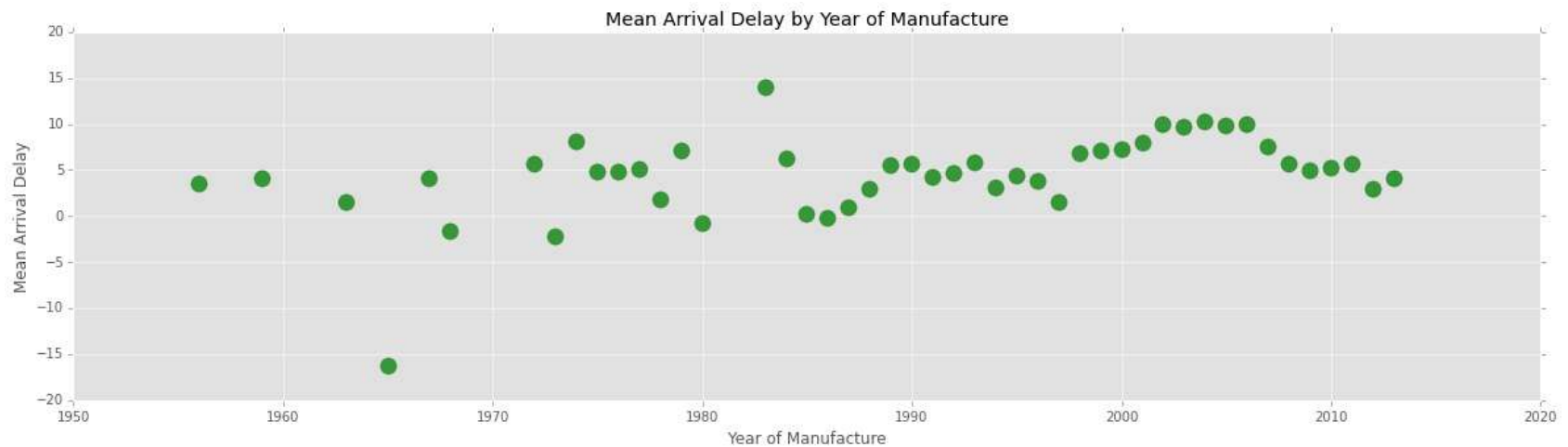


```
Out[310]: <ggplot: (316346325)>
```

The above plot shows how the mean departure delay for planes manufactured during different years. The older planes seem to have a lower mean departure delay which is quite strange. It could be because these planes are used for shorter distances and lesser number of journeys which results in lower delays as compared to the flights that were manufactured recently which might make a lot of journeys and hence have delays that might average out to be a higher number.

```
In [311]: #Group the data in the new merged data frame based on manufactured year and take the mean of the other columns
flights_planes_grouped = flights_planes.groupby('year_y').mean()
flights_planes_grouped["year_y"] = flights_planes_grouped.index
flights_planes_grouped.head(5)

#Create a plot that shows mean arrival delay by year of manufacture
gsize = theme_matplotlib(rc={"figure.figsize": "20, 5"}, matplotlib_defaults=False)
ggplot(aes(x="year_y",y="arr_delay"),data=flights_planes_grouped)+\
geom_point(color="green", size = 150, alpha = 0.75)+xlab("Year of Manufacture")+\\
ylab("Mean Arrival Delay")+ggtitle("Mean Arrival Delay by Year of Manufacture")+gsize
```



```
Out[311]: <ggplot: (344065509)>
```

Similar to the plot of the departure delay spread out by year of manufacture, the above plot shows how the mean arrival delay for planes manufactured during different years. The spread of the delay is quite consistent but there is a slightly lesser delay in the planes manufactured in the earlier years.

```
In [ ]:
```